

are unlikely to explain the specific effect identified by our empirical design.

5.4.1 Interest Rates

A natural concern is that other contemporaneous shocks may have disproportionately affected labor demand for junior workers in GenAI-adopting firms relative to non-adopters. The first such concern is that GenAI adopters may be more sensitive to monetary policy cycles and thus may have reacted more strongly to the 2022–2023 interest rate hikes. However, several pieces of evidence suggest that this mechanism does not drive our findings. First, if monetary policy shocks had played a role, we would expect to see effects in the pre-trend coefficients of Figure 7, yet none are observed. Second, even if interest rate changes disproportionately affected industries where adopters are concentrated or had stronger impacts on junior roles, these dynamics should be absorbed by the industry-by-seniority-by-time fixed effects in the triple-difference estimates in Section 5.2.

Third, we directly test for this channel following Ottonello and Winberry (2020). We use the high-frequency identified monetary-policy surprises of Jarocinski and Karadi (2020), aggregated to the quarterly level as the in-quarter sum of monthly surprises, and estimate the differential-sensitivity regression

$$\log L_{i,t+1} - \log L_{i,t} = \alpha_i + \gamma_t + \beta(\text{Adopter}_i \times \varepsilon_t^m) + \eta_{i,t}, \quad (14)$$

on the pre-2022 firm-quarter panel ($t \in 2015\text{Q1} - 2021\text{Q3}$), to see if our adopter firms were disproportionately responsive to previous monetary policy shocks. The outcome is the forward one-quarter log change in total or junior employment, α_i and γ_t are firm and year-quarter fixed effects. Under the monetary-policy confounder story, $\beta < 0$: adopters should contract *more* than non-adopters in the quarter following a tightening shock.

Table 3 reports the estimates. For total employment, the interaction is small and of the *opposite* sign to what the confounder story would predict ($\hat{\beta} = 0.046$, s.e. 0.020): adopters expand slightly more, not less, than non-adopters in the quarter following a tightening. For junior employment, the interaction is small and statistically insignificant ($\hat{\beta} = 0.018$, s.e. 0.025). Either way, there is no evidence that adopters contract differentially after identified monetary-policy shocks. Appendix A.23 reports lag-augmented local-projection impulse responses (Jordà, 2005; Montiel Olea and Plagborg-Møller, 2021) across horizons

Table 3: Sensitivity of adopter employment to identified monetary-policy shocks

	$\log L_{i,t+1} - \log L_{i,t}$	
	Total employment	Junior employment
Adopter _{<i>i</i>} $\times$ ε_t^m	0.046** (0.020)	0.018 (0.025)
Firm FE, Year-quarter FE	Yes	Yes
Firms	275,809	272,549
Observations	5,919,703	5,692,315
Quarters	23	23

Notes. Shock quarters $t \in 2015Q1\text{--}2021Q3$ (sample stops before 2022 so the forward one-quarter outcome window closes no later than 2021Q4). The shock ε_t^m is the in-quarter sum of monthly pure monetary-policy surprises from Jarocinski and Karadi (2020). Each column is one regression of the forward one-quarter log change in total (column 1) or juniors (column 2) employment on Adopter_{*i*} $\times$ ε_t^m , with firm and year-quarter fixed effects. Outcomes are winsorised at 0.5/99.5%. Standard errors in parentheses are Cameron–Gelbach–Miller two-way cluster-robust at the firm and year-quarter level; *t*-statistics use $\min(G_{\text{firm}}, G_{\text{quarter}}) - 1$ degrees of freedom. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

$h = 0, \dots, 8$ quarters, which deliver the same null at every horizon. This finding is also consistent with evidence in the literature showing that larger firms, which are more likely to adopt GenAI, tend to be less sensitive to monetary-policy shocks (see Chodorow-Reich, 2014; Gertler and Gilchrist, 1994).

5.4.2 Post-COVID Dynamics, Remote Work, and Labor Supply Shocks

Beyond monetary policy, three interrelated concerns arise from the unusual labor-market dynamics of the post-pandemic period, which overlaps with the diffusion of GenAI.

Post-COVID dynamics. A potential confounder is the post-COVID hiring boom in the technology sector (U.S. Bureau of Labor Statistics, 2025). If this boom was followed by a correction, it could disproportionately affect GenAI adopters, who are more heavily represented in the Information sector (Section 4.2). However, these dynamics are also likely to be absorbed by the industry-by-seniority-by-time fixed effects. Moreover, if boom-bust dynamics were driving our results, we would expect to observe a significant relative increase in junior employment among adopters (relative to non-adopters) prior to GenAI adoption, which we do not (see Figures 4 and 7).

Remote work. Another potential confounder is the post-pandemic normalization of remote work. Work-from-home arrangements may reduce demand for junior workers for reasons unrelated to automation, for example by increasing supervision costs and weakening opportunities for on-the-job learning. This concern is particularly relevant because occupations that are well suited to remote work overlap substantially with those most exposed to GenAI (Lambert and Schindler, 2026). To assess this possibility, we re-estimate our seniority triple-difference specification (Equation 11) separately using employment in positions with high and low remote suitability, based on the position-level `remote_suitability` score in the Revelio individual-positions data.³⁰ The seniority-biased decline in junior employment at adopting firms appears in both groups and is similar in magnitude across them. In particular, we continue to find a substantial junior decline among low-remote-suitability positions, which are unlikely to be performed from home. These results suggest that our findings are not driven by the post-pandemic normalization of remote work. Appendix A.24 provides additional details on the data, methodology, and results, including the corresponding triple-difference estimates in Figure A.26.

Labor supply shocks. A final concern is reverse causality—namely, that firms adopting GenAI were disproportionately affected by negative supply shocks to junior labor, which prompted adoption. To assess this possibility, we examine a more direct measure of labor demand: job postings. Since the job postings data do not include seniority information, this analysis captures total demand for workers rather than its distribution by seniority, and the job postings data begin only in 2021Q4. Figure A.27, reported in Appendix A.25, presents the results: starting in 2023Q1, the number of job postings declined significantly in adopting firms relative to non-adopters. This provides suggestive evidence that the observed decline in junior employment reflects a contraction in labor demand rather than a supply-driven shock.

³⁰More details on Revelio’s remote-suitability classification methodology are available at <https://www.reveliolabs.com/news/business/which-companies-will-succeed-in-the-new-world-of-remote-work/>.

5.5 Decomposing the Decline in Junior Employment into Flows

The decline in junior employment in adopting firms can arise through three channels: (i) reduced hiring, (ii) increased separations, or (iii) increased promotions to senior positions. Our LinkedIn-based résumé data can be viewed as a detailed matched employer-employee dataset, allowing us to track these worker flows over time. Specifically, we estimate separate DiD regressions of the following form:

$$y_{it} = \alpha + \beta (\text{Adopt}_i \times \text{Post}_t) + \delta_t + \gamma_i + \xi_{pt} + \varepsilon_{it}, \quad (15)$$

where y_{it} denotes the number of junior hires, number of junior separations, or the junior promotion rate³¹ in firm i at quarter t . γ_i denotes firm fixed effects, δ_t are time fixed effects and ξ_{pt} are industry-by-time fixed effects.

Table 4 reports the estimated β from Equation 15. The results indicate that the sharp contraction in junior employment among adopters is driven primarily by a slowdown in hiring. Specifically, the coefficient on *Hiring* implies that, relative to non-adopters, GenAI-adopting firms hired on average 4.0 fewer junior workers per quarter after 2023Q1—about an 80 percent reduction relative to the pre-period mean of 5.06 hires per firm-quarter. Junior separations also declined among adopters, by roughly 1.1 workers per quarter, partially offsetting the hiring contraction; but this reduction is about one-fourth the magnitude of the hiring decline, so net junior employment still contracts sharply.³² Junior promotion rates rose by a small but statistically significant 0.033 percentage points (relative to a pre-period mean of 0.82 percent). Because promotions move workers out of the junior pool, this rise further reduces the junior stock, though its magnitude is very small relative to the hiring channel. Appendix A.28 explores sectoral heterogeneity in the decline in junior hiring, showing that the contraction is broad-based across industries and not driven disproportionately by any single sector.

³¹The junior promotion rate is defined as $\frac{\text{Promotions}_t}{\text{Junior Employment}_{t-1}}$, expressed in percentage points.

³²See Supplemental Appendix A.26 for the time-series DiD results for junior hires and exits.